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Konstantin Volodin

Applied ML & Data Science

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SUMMARY

*Data Scientist with 5+ years of experience. Owner of the TRV platform at IRCC - managed the full stack from PySpark data pipelines to AWS deployment. Redesigned the core classification model, cutting false approvals by **54%**. Background in operations research, optimization, and a Masters in Analytics from McGill. Looking for senior DS/MLE roles in Montreal where I own platform architecture and drive technical decisions.*

SKILLS

Programming	Python • SQL • R • JavaScript
ML & Data	PySpark • Pandas • PyArrow • PyTorch • HuggingFace • Gurobi • Plotly • PowerBI
Platforms	AWS (EC2-S3-ECS-ECR-CDK) • Docker (Compose) • Git(Hub Actions) • Azure DevOps • Linux • Databricks
Languages	English (Fluent) • Russian (Fluent) • French (B1)

EXPERIENCE

Data Scientist, *Immigration, Refugees and Citizenship Canada* — Oct 2022 - Present
Advanced Analytics Solutions Center

- Lead engineer for the Temporary Resident Visa, the team's highest-throughput ML platform, **processing ~150K files weekly** through production PySpark, PyArrow, and Pandas pipelines over Parquet storage with real-time decision support.
- Redesigned a high-risk classification model with novel feature engineering, **reducing false approvals by 54%**.
- Optimized a production PySpark pipeline end-to-end, **reducing runtime 60% (8h to 3h)** through Spark job refactoring, deduplication, and compute environment tuning.
- Architected a 40+ SQL star schema feature layer, transforming raw operational data into model-ready datasets, powering a **30+ bin classification system** used by hundreds of IRCC officers.
- Coordinated the full ML deployment pipeline: Docker containers, published to AWS ECR, ECS via CDK, with PR-tracked versions in Azure DevOps and structured logging for monitoring.
- Expanded platform coverage to new applicant categories: cross-model integrations and initiatives like FIFA 2026.

Data Analyst, *Immigration, Refugees and Citizenship Canada* — Dec 2020 - Oct 2022
Anti Racism Task Force

- Technical lead for a cross-departmental team
- Built R-Shiny dashboards and performed ad-hoc analyses on an 8,000 employee dataset for equity gaps.
- Designed a department-wide survey; presented findings to DG-level executives to inform policy decisions.

Research Assistant - Capstone, *CAE Inc.* — Oct 2022 - Apr 2023

- Built a forecasting pipeline with model selection to predict demand for thousands of aviation components.
- Designed an inventory policy balancing demand, lead times, and service-level targets to reduce inventory costs.

Research Assistant, *University of Ottawa* — May 2019 - Aug 2019

- Developed a mixed-integer scheduling model with 100+ constraints to generate a year-long timetable, cutting planning time from weeks to hours and improving fairness and utilization.
- Built a web app with React and Django to let staff run and evaluate scheduling scenarios.

PROJECTS

Homelab, *Self-Hosted Infrastructure* — 2026 - Present

- Multi-node Docker Compose stack (media, recipes, photos, books) with ARR automation and infrastructure as code approach.
- Secured with authelia SSO, Caddy reverse proxy with internal CA, Tailscale mesh VPN, custom domain (apps.voxlab.home).

Infinite DnD, *Multi-Agent AI Storytelling* — 2025 - Present

- Autonomous D&D; game engine with multi-agent AI running the sessions - world-building, dialogue, encounter.
- Python, PydanticAI, llama.cpp for fully local inference on consumer hardware.

EDUCATION

Master of Management in Analytics, *McGill University* — 2022 - 2023
Courses: Deep Learning, Statistical Modelling, NLP, Data Visualization, Optimization

Bachelor of Commerce (Finance), *University of Ottawa* — 2016 - 2020
Courses: Mathematical Optimization, Simulation Analytics, Statistics, Financial Modelling